Neural Networks 201 (2026) 108950

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

CC-Net: A cross-hierarchical context-aware network for medical image segmentation

Xiaoyan Zhang[a], Zheng Zhao[a], Weiqiang Sun[a], Yongqin Zhang[b], Chunlin Yu[c], Xiangfu Meng[a,*]

[a] School of Electronics and Information Engineering, Liaoning Technical University, Huludao, 125105, Liaoning, China
[b] School of Computer Science and Technology, Hainan University, Haikou, 570208, Hainan, China
[c] School of Computer and Communication Engineering, University of Science and Technology Beijing, 100083, Beijing, China

ARTICLE INFO

Keywords:
Hierarchical interaction
Medical image segmentation
Cross-hierarchical
Context-aware

ABSTRACT

Accurate segmentation of lesion regions in medical images is crucial for early diagnosis and precise treatment of diseases. Existing U-shaped segmentation networks typically employ direct connections between features of the same hierarchical level in the encoder-decoder structure; however, the lack of explicit interaction between different hierarchical features limits the effective integration of shallow structural information and deep semantic features, resulting in challenges when delineating complex boundaries and fine-grained structures. To address these challenges, we propose a Cross-Hierarchical Context-Aware Network (CC-Net). The model comprises four key components: The Cross-Hierarchical Feature Aggregation (CFA) Module aggregates shallow and deep features separately, explicitly modeling semantic relationships across hierarchical levels. The Global Feature Aggregation (GFA) Module integrates features from multiple levels to construct a unified global representation, providing multi-level semantic context during cross-branch feature interactions. The Cross-Branch Semantic Supplement (CSS) Module injects the global semantic information obtained from GFA into the CFA-aggregated shallow and deep features, enhancing overall structural awareness. The Enhanced Feature (EF) Module strengthens the model's discriminative capability for target regions while effectively suppressing background noise. Extensive experiments on six public medical image datasets demonstrate that CC-Net consistently outperforms state-of-the-art segmentation methods across all evaluation metrics. The code is available at https://github.com/zz0226zz/CC-Net.

1. Introduction

In medical image analysis, accurate lesion segmentation remains a fundamental yet crucial task. From early screening and computer-assisted diagnosis to the formulation of subsequent treatment strategies, each stage relies heavily on precise delineation of lesion boundaries; accordingly, segmentation quality exerts a substantial influence on clinicians' overall assessment of disease status as well as on the accuracy and efficiency of medical decision-making (Azad et al., 2024). The advent of Fully Convolutional Networks (Long et al., 2015) made end-to-end pixel-wise prediction feasible and, in doing so, marked a major advance in semantic segmentation. Building upon this foundation, U-Net effectively alleviated the loss of fine-grained details during downsampling through the introduction of skip connections, thereby demonstrating particularly strong performance in medical image segmentation tasks. Since then, convolutional neural network architectures developed around the U-shaped paradigm have continued to evolve, with representative models such as U-Net (Ronneberger et al., 2015) and UNet++ (Zhou et al., 2018) being widely adopted in related studies (Fan et al., 2020; Huang et al., 2020; Jha et al., 2020; Zhang et al., 2018). In general, these methods enhance the representation of target regions by integrating multi-level feature information and strengthening semantic interaction between the encoder and decoder via skip connections, ultimately enabling more precise localization and segmentation.

Although CNNs possess a clear advantage in local texture extraction, their limitations are equally apparent. During encoding, they often struggle to model global semantic information adequately; during decoding, critical low-level spatial details are also prone to incomplete recovery. For this reason, attention mechanisms have been increasingly incorporated into segmentation models to strengthen global context perception and semantic modeling. Representative methods, such as Swin-Unet (Cao et al., 2022), MISSFormer (Huang et al., 2022), and

* Corresponding author.
E-mail addresses: zhangxiaoyan@lntu.edu.cn (X. Zhang), zhaozheng0226@126.com (Z. Zhao), lnswq17@126.com (W. Sun), zyq1815@163.com (Y. Zhang), yuchunlin627@126.com (C. Yu), mengxiangfu@lntu.edu.cn (X. Meng).

https://doi.org/10.1016/j.neunet.2026.108950
Received 27 October 2025; Received in revised form 18 March 2026; Accepted 4 April 2026
Available online 10 April 2026
0893-6080/

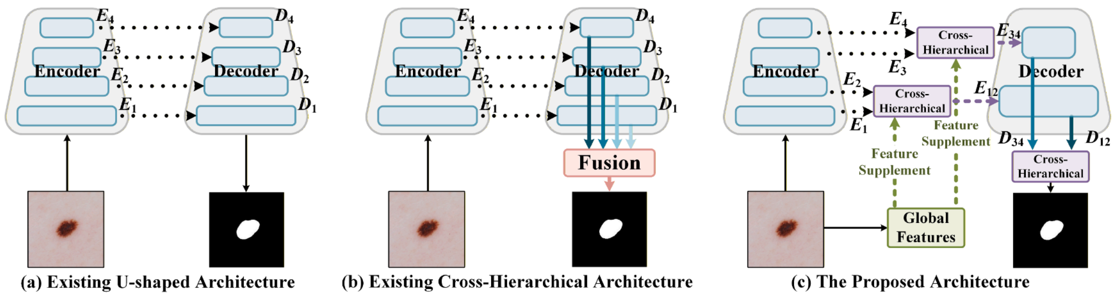

Fig. 1. Three distinct architectures for medical image segmentation: (a) existing u-shaped architecture; (b) existing cross-hierarchical architecture; (c) the proposed architecture.

MTU-Net (Wu et al., 2023), have all demonstrated notable improvements in global semantic understanding. At the same time, global modeling alone is often insufficient to preserve fine-grained local details. This has led many studies to further explore hybrid architectures that combine CNNs with Transformers, such as MiXFormer (Liu et al., 2024a), LW-CTrans (Kuang et al., 2025), and EFFM (Zhang et al., 2026). To a considerable extent, these frameworks capitalize on the complementary strengths of both paradigms: CNNs are well suited for extracting fine-grained textures and local structures, whereas Transformers excel at modeling long-range dependencies and capturing global semantic representations. Through this complementary integration, models can obtain richer multi-scale and multi-level feature representations. Compared with single-architecture designs, such hybrids generally improve not only segmentation accuracy and robustness, but also adaptability to complex structures and boundary regions. Beyond these approaches, several Mamba-based models have also shown considerable promise in recent years, including MambaDiff (Liu et al., 2025), PMGNet (Zhu et al., 2025), Swin-UMamba † (Liu et al., 2024b), MultiSegMamba (Lumetti et al., 2025), and Zhu et al. (2023), which incorporates a boundary detection mechanism. Overall, by further extending contextual modeling capacity, these methods enhance the refinement of segmentation results while also improving model robustness and generalization to a certain extent.

Although existing U-shaped and hybrid architectures have achieved considerable progress in complex structure segmentation, their limitations remain evident, particularly in fine-grained scenarios. Prior studies have shown that features at different hierarchical levels exhibit pronounced semantic disparities: shallow features tend to preserve fine-grained details and contours, whereas deep features are more effective at representing overall structures and high-level semantic information (Zeiler & Fergus, 2014). Nevertheless, conventional U-shaped architectures and their variants generally treat features from different levels as independent representations (as illustrated in Fig. 1(a)), with insufficient modeling of continuity and complementarity across hierarchies. In the context of complex lesion and fine-structure segmentation, such a design can easily lead to inadequate coordination between shallow structural cues and deep semantic information, thereby weakening the model's ability to precisely delineate blurred boundaries, small-scale lesions, and morphologically complex regions. Representative methods further illustrate this issue. Although TransFuse (Zhang et al., 2021) and EMCAD (Rahman et al., 2024) have introduced improved feature fusion strategies, their fusion processes still operate predominantly within the same hierarchical level, leaving cross-level complementary information insufficiently exploited. As a result, when confronted with multi-scale structures or regions with ambiguous boundaries, these methods often struggle to effectively coordinate shallow structural information with deep semantic cues, which in turn constrains segmentation precision. CASCADE (Rahman & Marculescu, 2023), by contrast, attempts to aggregate features generated from different decoder layers only at the final stage to produce predictions (as shown in Fig. 1(b)). Yet because this fusion occurs only at the end of decoding, the contribution of cross-level information is introduced relatively late. Without continuous and collaborative optimization of shallow structural features and deep semantic features throughout the entire encoding-decoding process, the model's segmentation accuracy for small lesions, blurred boundaries, and morphologically complex regions still leaves substantial room for improvement.

To address the aforementioned challenges, we propose a hybrid cross-hierarchical feature modeling mechanism, as illustrated in Fig. 1(c), to explicitly characterize semantic dependencies across different feature levels. In the Transformer-dominant encoder branch, the first two shallow layers are fused in a cross-hierarchical manner to more effectively preserve fine-grained details and contour information; correspondingly, the last two deep layers are further integrated to strengthen the representation of overall structures and high-level semantic information. In parallel, within the CNN auxiliary encoder branch, multi-level features are first aggregated into a unified global representation and then refined through a cross-branch feature supplementation strategy. For shallow features, this strategy primarily enhances structural consistency and spatial awareness; for deep features, it further introduces global contextual information while maintaining sensitivity to local details, thereby enabling more discriminative modeling of high-level semantic features.

Building upon the aforementioned hybrid cross-hierarchical feature modeling framework, this study introduces the Cross-Hierarchical Context-Aware Network (CC-Net). Its overall architecture comprises two parallel feature extraction branches: a Pyramid Vision Transformer (PVT) (Wang et al., 2021) branch and a convolutional branch. Together, these branches encompass four pivotal modules, each dedicated to specific feature modeling objectives, uniquely indispensable and functionally distinct. Within the PVT branch, the Cross-Hierarchical Feature Aggregation (CFA) module separately consolidates shallow and deep features, explicitly capturing semantic correlations across hierarchical levels and enhancing the synergistic representation of low-level structural cues and high-level semantic information. In the convolutional branch, the Global Feature Aggregation (GFA) module integrates multi-scale features from various layers into a unified global representation, subsequently providing overarching guidance for cross-branch feature enhancement. The Enhanced Feature (EF) module performs self-augmentation within each branch, simultaneously reinforcing target-relevant signals and suppressing background noise, thereby rendering single-branch feature representations more robust and discriminative. The Cross-Branch Semantic Supplement (CSS) module employs a cross-branch feature transfer strategy, transmitting global features aggregated by GFA from the convolutional branch to the PVT branch, enriching both shallow and deep features with multi-level semantic information while augmenting structural awareness. Owing to the complementary functionality of these four modules, both within and across branches, CC-Net effectively preserves fine-grained details while simultaneously strengthening global contextual understanding. This architectural synergy markedly enhances segmentation performance, particularly in scenarios involving complex lesions and intricate anatomical structures.

The contributions of this work are as follows:

(1) We propose the CFA module to jointly model shallow and deep features while explicitly characterizing semantic correlations across different hierarchical levels. After multi-stage feature processing, the shallow and deep features are further fused through this module, thereby providing more comprehensive and consistent representational support for the final prediction.

(2) We design the GFA module and the CSS module. Specifically, GFA integrates features from different hierarchical levels into a unified global representation; based on this, CSS injects the global representation into the shallow and deep features aggregated by CFA, enriching them with global semantic context while enhancing structural awareness.

(3) We propose the EF module to further improve the model's ability to identify target regions. By reinforcing key target-related information while effectively suppressing background noise, this module enhances the discriminative power of feature representations.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 presents the proposed method in detail. Section 4 reports performance comparisons, ablation studies, and parameter analyses on six medical image datasets. Section 5 discusses the limitations of the proposed approach and outlines directions for future research. Finally, Section 6 concludes the paper.

2. Related work

2.1. Medical image segmentation methods based on U-shaped architectures

The central objective of medical image segmentation is the accurate identification of critical anatomical structures, including lesions and pathological regions. This process is not only closely tied to the reliability of early screening and computer-assisted diagnosis, but also provides an essential basis for subsequent treatment planning; as such, it substantially influences the accuracy and efficiency of clinical assessment. Owing to its clear structural design, architectural simplicity, and stable performance, the U-shaped architecture has gradually become one of the most representative foundational frameworks in medical image segmentation (Chen et al., 2022; Ibtehaz and Kihara, 2023; Isensee et al., 2021; Jiang et al., 2025; Zhang et al., 2021). Broadly speaking, this architecture consists of three principal components: an encoder that progressively extracts image features while compressing spatial information, a decoder that incrementally restores spatial resolution to produce segmentation outputs, and skip connections that transmit multi-scale features between the encoder and decoder, thereby preserving local details as much as possible and improving segmentation accuracy.

To further improve the performance of U-Net-based models, researchers have explored a variety of enhancement strategies targeting key architectural components, including the encoder, skip connections, decoder, and attention mechanisms. Representative examples illustrate this trend clearly. EMCAD (Rahman et al., 2024) introduces an efficient feature decoder that markedly improves segmentation performance while reducing computational overhead. CASCADE (Rahman & Marculescu, 2023), through a cascaded attention-based decoder, effectively integrates multi-scale semantic information from hierarchical vision Transformers, enabling the model to better balance fine-grained details with global contextual understanding. MiXFormer (Liu et al., 2024a) goes a step further by designing a hybrid encoder that combines CNNs and Transformers, allowing local feature extraction and global contextual modeling to operate in a complementary and synergistic manner.

Taken together, these U-Net-based variants, through continual architectural refinement, achieve a favorable balance among segmentation accuracy, computational efficiency, and multi-scale feature representation, thereby providing a solid foundation for more precise and intelligent computer-aided diagnosis.

2.2. Medical image segmentation methods employing non-U-shaped architectures

With the continued deepening of related research, an increasing number of non-U-shaped architectures have been introduced into medical image segmentation in an effort to overcome the inherent limitations of conventional U-Net in feature representation and contextual modeling. Unlike approaches that adhere to the classical encoder-decoder paradigm, these methods are often redesigned from the perspectives of network organization, information interaction, and task formulation. MedT (Valanarasu et al., 2021), for instance, adopts a dual-branch architecture to extract global features from the entire image and local features from image patches, thereby enabling complementary multi-scale feature fusion. I^2U-Net (Dai et al., 2024), by contrast, facilitates more thorough information exchange between dual paths, allowing historical features to be efficiently reused and more deeply exploited, which in turn yields more comprehensive feature representations. Beyond static image segmentation, several methods have extended their modeling capacity to more complex dynamic scenarios. Within a diffusion-model framework, Diff-VPS (Lu et al., 2024) introduces multi-task supervision and temporal reasoning modules to characterize the highly camouflaged and temporally redundant nature of polyps in video data, leading to a marked improvement in dynamic segmentation accuracy. RCPCL (Huang et al., 2025) combines cross copy-paste with uncertainty-constrained strategies to achieve a more balanced trade-off between synthetic label quality and perturbation control, thereby further unlocking the potential of consistency learning. Meanwhile, MCANet (Shao et al., 2025), built upon a multi-scale cross-axis attention mechanism, effectively integrates global and spatial features and, through a unified decoder, achieves precise segmentation of lesions with diverse morphologies. S2VNet (Ding et al., 2024), centered on a clustering-based slice-to-volume propagation strategy, uses the clustering results of preceding slices to guide the segmentation of the current slice, substantially improving both efficiency and generalization while accommodating automatic and interactive tasks within a unified framework.

Overall, these non-U-shaped models transcend the limitations of traditional architectures from multiple perspectives. Whether through the coordinated integration of local and global information, the introduction of temporal modeling and multi-task supervision, or the use of clustering-guided strategies, they continue to expand the modeling frontier of medical image segmentation and drive the field toward greater precision and stronger adaptability.

3. Methods

Fig. 2 presents the overall architecture of CC-Net. In this study, PVT is adopted as the backbone, and its design is inspired, to some extent, by CASCADE (Rahman & Marculescu, 2023) and EMCAD (Rahman et al., 2024). Given the strong performance of these two methods in related tasks, we retain the corresponding backbone design and further construct a feature modeling framework tailored to our task.

During feature extraction, the input medical image is fed simultaneously into the PVT branch and the convolutional branch for parallel processing. The PVT branch serves as the primary encoder and extracts four hierarchical feature representations, denoted as $E_i \in \mathbb{R}^{\frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times C_i}$, where $C_i \in 64, 128, 320, 512$ and $i \in 1, 2, 3, 4$. For the shallow part, features E_1 and E_2 are fused through the CFA module to enhance the representation of fine details and contour information; for the deep part, E_3 and E_4 are likewise aggregated by CFA, thereby strengthening the modeling of high-level semantic information. Meanwhile, the convolutional branch further extracts three additional features, denoted as $L_j \in \mathbb{R}^{\frac{H}{2^{j+2}} \times \frac{W}{2^{j+2}} \times C_j}$, where $C_j \in 128, 320, 512$ and $j \in 1, 2, 3$. These features are subsequently integrated into a unified global representation through the GFA module and then introduced into the PVT branch via the CSS module, so as to supplement both shallow and deep features

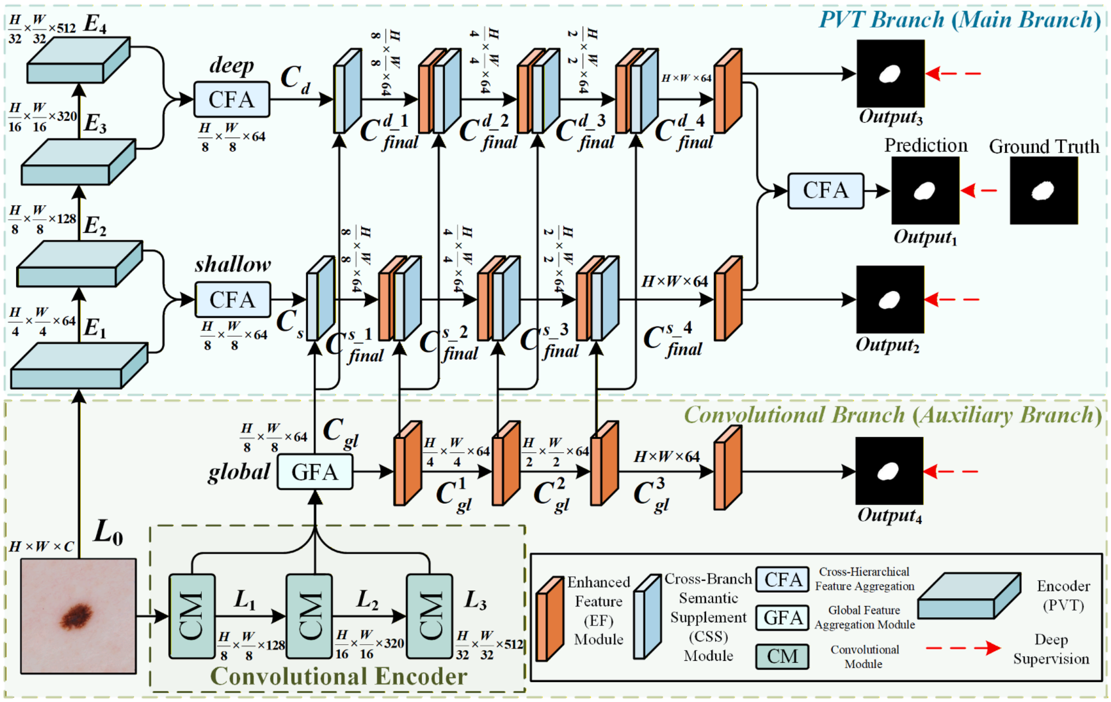

Fig. 2. The overall architecture of CC-Net comprises two primary branches: a PVT branch and a convolutional branch, and incorporates four key submodules: the CFA module, the GFA module, the EF module, and the CSS module.

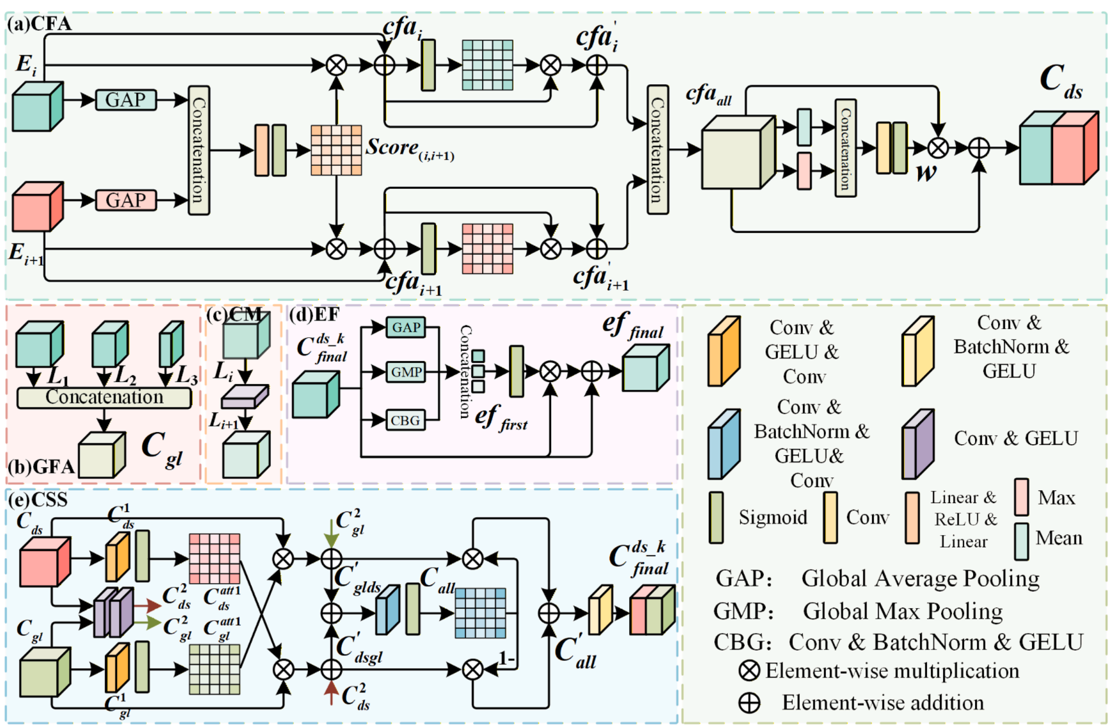

Fig. 3. The five principal sub-modules of CC-Net: **(a)** CFA Module; **(b)** GFA Module; **(c)** CM; **(d)** EF Module; **(e)** CSS Module.

with global semantic information. At the same time, the EF module further emphasizes target-relevant information while suppressing background noise. After multiple rounds of feature refinement, the updated shallow and deep features are fused once again through the CFA module, ultimately producing a prediction map with more precise boundaries and clearer structural delineation.

3.1. Cross-hierarchical feature aggregation module

Features at different hierarchical levels exhibit distinct semantic characteristics: shallow features focus more on fine-grained details and contour information, while deep features are better at capturing global structures and high-level semantic information (Zeiler & Fergus, 2014).

To effectively integrate the complementary information across these levels, we propose the CFA module, whose architecture is illustrated in Fig. 3(a).

To enhance semantic consistency between features at different hierarchical levels, we first apply GAP to the two input features, E_i and E_{i+1}, respectively, in order to extract the global response of each channel. The resulting features are then concatenated along the channel dimension to form a joint representation that simultaneously captures the global semantic information from adjacent layers. To fully integrate cross-channel information and explicitly model the nonlinear dependencies among different channels, a multi-layer perceptron is employed to map the joint representation. Finally, attention weights are generated through a Sigmoid activation function. The computation is formulated as follows:

$$Score_{(i,i+1)} = \sigma(\mathrm{MLP}(\mathrm{Concat}(\mathrm{GAP}(E_i), \mathrm{GAP}(E_{i+1})))), \tag{1}$$

where $\mathrm{GAP}(\cdot)$ denotes global average pooling, Concat denotes concatenation along the channel dimension, $\mathrm{MLP}(\cdot)$ represents a multi-layer perceptron, and $\sigma(\cdot)$ denotes the Sigmoid activation function. $Score_{(i,i+1)}$ represents the final attention weight. The purpose of this operation is to enhance the shared key semantic responses between the two features, thereby achieving cross-feature semantic consistency.

Subsequently, the attention weights are applied to the original features E_i and E_{i+1} through element-wise multiplication, followed by residual fusion with the corresponding original features. Since the attention weights are generated from the joint information of both features, this process introduces complementary semantic cues during feature modulation, thereby reducing the semantic discrepancy between adjacent features and improving the consistency of their representations. The computation process is formulated as follows:

$$\begin{cases} cfa_i = Score_{(i,i+1)} \times E_i + E_i, \\ cfa_{i+1} = Score_{(i,i+1)} \times E_{i+1} + E_{i+1}, \end{cases} \tag{2}$$

where cfa_i and cfa_{i+1} denote the output features after attention modulation and residual fusion. Then, a second self-enhancement step is applied to each feature individually to further reinforce the commonly important response regions. Finally, the enhanced features are concatenated along the channel dimension to preserve complementary and consistent representations. The computation is formulated as follows:

$$\begin{cases} cfa_i' = \sigma(cfa_i) \times cfa_i + cfa_i, \\ cfa_{i+1}' = \sigma(cfa_{i+1}) \times cfa_{i+1} + cfa_{i+1}, \\ cfa_{all} = \mathrm{Concat}(cfa_i', cfa_{i+1}'), \end{cases} \tag{3}$$

where cfa_i' and cfa_{i+1}' denote the output features after the second attention enhancement, and cfa_{all} represents the fused feature obtained by concatenating them along the channel dimension. To further improve cross-feature semantic consistency, max pooling and average pooling are applied to cfa_{all} to capture its spatial responses from different statistical perspectives. The pooled features are concatenated and compressed by a 1×1 convolution, followed by a Sigmoid activation to generate the attention weights. The weights are then applied to the original fused features through element-wise multiplication and residual addition, yielding the optimized feature representation. The computation can be formulated as follows:

$$\begin{cases} w = \sigma(\mathrm{Conv}_{1\times1}(\mathrm{Concat}(\max(cfa_{all}), \mathrm{avg}(cfa_{all})))), \\ C_{ds} = w \times cfa_{all} + cfa_{all}, \end{cases} \tag{4}$$

where $\max(\cdot)$ denotes the max pooling operation, $\mathrm{avg}(\cdot)$ denotes the average pooling operation, w represents the final attention weight, and C_{ds} denotes the final output feature of the module, with its channel dimension compressed to match that of the input. Through this series of operations, the network adaptively strengthens key response regions that are semantically consistent and discriminative across features while preserving the original structural information, thereby effectively enhancing the semantic consistency and discriminative capacity of the feature representation.

3.2. Global feature aggregation module

Single-level features often struggle to simultaneously capture fine-grained local details and comprehensive global semantics, thereby limiting the model's ability to accurately interpret and segment complex structures. To address this, we propose a GFA Module (as shown in Fig. 3(b)) designed to effectively fuse multi-level features into a unified representation enriched with semantic information.

First, the initial feature L_0 is fed into the Convolutional Module (CM), where three successive rounds of feature extraction generate three hierarchical semantic feature maps. The detailed architecture of CM is illustrated in Fig. 3(c), mainly consisting of convolutional layers and Group Normalization (GN). In particular, GN is introduced to regulate the feature distributions across different levels, reduce statistical discrepancies among multi-level features, and thereby provide more stable representations for subsequent feature aggregation. The computational process can be formulated as follows:

$$L_{i+1} = \mathrm{GN}(\mathrm{Conv}_{3\times3}(L_i)), i \in \{0, 1, 2\}, \tag{5}$$

where $\mathrm{Conv}_{3\times3}(\cdot)$ denotes a 3×3 convolution operation, $\mathrm{GN}(\cdot)$ represents Group Normalization, and L_{i+1} is the output feature after this operation. The above operation is performed three times to progressively extract hierarchical semantic features. Subsequently, the three processed feature maps, L_1, L_2, and L_3, are resized to a consistent spatial resolution and concatenated along the channel dimension to construct a unified feature representation. Since this representation integrates semantic information from multiple hierarchical levels within the convolutional branch-spanning from low-level to high-level features-it enables comprehensive modeling of the branch's semantic information along the feature hierarchy. Therefore, we define this as the global feature representation. It is important to note that the term "global" here does not refer to spatially global modeling, but rather emphasizes the unified aggregation of all semantic information across the feature hierarchy of the branch. The computation is formulated as follows:

$$C_{gl} = \mathrm{Conv}_{1\times1}(\mathrm{Concat}(L_1, \mathrm{UP}^2(L_2), \mathrm{UP}^4(L_3))), \tag{6}$$

where $\mathrm{Concat}(\cdot)$ denotes the concatenation of multi-level features along the channel dimension, $\mathrm{UP}^k(\cdot)$ represents upsampling the feature by a factor of k, and $\mathrm{Conv}_{1\times1}(\cdot)$ denotes a 1×1 convolution operation, which is primarily employed to compress the channel dimension of the concatenated features, thereby producing a feature representation with unified dimensionality, and C_{gl} is the final output of this module. Finally, the resulting global feature serves as a cross-branch semantic prior and supplements the shallow and deep features of the PVT branch, which have been aggregated via the CFA module, through the CSS module, thereby enhancing the overall semantic understanding of the main branch features.

3.3. Enhanced feature module

The EF module is designed to further improve the model's ability to identify target regions by emphasizing salient features while suppressing redundant or irrelevant background information, thereby enhancing segmentation accuracy. As illustrated in Fig. 3(d), the module operates in two stages: a feature extraction stage and a noise filtering stage.

During the feature extraction stage, the input features $C_{final}^{ds\_k}, k \in \{1, 2, 3, 4\}$ are processed in parallel through three distinct operations: Global Average Pooling (GAP), Global Max Pooling (GMP), and Convolution-BatchNorm-GELU (CBG). Specifically, GAP computes the mean activation of each channel across all spatial positions, capturing the global spatial distribution of the features. In contrast, GMP selects the maximum activation within each channel, highlighting the most discriminative regions and thereby reinforcing the representation of salient target areas. The CBG operation applies local convolution to model spatial relationships within neighborhoods, effectively capturing fine-grained structures and local details. The outputs from these three operations are subsequently concatenated along the channel dimension, enabling a comprehensive multi-perspective representation of the feature information, which can be formalized as:

$$
\begin{aligned}
ef_{\text{first}} = \text{Concat}(&\text{GAP}(C^{ds\_k}_{final}),\\
&\text{GMP}(C^{ds\_k}_{final}),\\
&\text{CBG}(C^{ds\_k}_{final})),\\
&k \in \{1,2,3,4\},
\end{aligned}
\tag{7}
$$

where GAP(·) denotes Global Average Pooling, GMP(·) denotes Global Max Pooling, CBG(·) denotes the Convolution-BatchNorm-GELU operation, and e_{first} refers to the feature produced by the above operations. The concatenated features then enter the noise filtering stage, where key information is enhanced and redundant noise is suppressed. Specifically, the concatenated features are first passed through a 1×1 convolution to reduce the channel dimension to match that of the original input features, followed by a Sigmoid activation to generate attention weights. These weights are applied to the input features via element-wise multiplication, enabling adaptive feature modulation: higher weights are assigned to salient features to "emphasize useful information," while lower weights are applied to redundant or irrelevant background features to "suppress noise." Finally, the weighted features are added residually to the original input features to obtain the enhanced feature representation. This process can be formally expressed as:

$$
ef_{final} = \sigma(\text{Conv}_{1\times1}(ef_{first})) \times C^{ds\_k}_{final} + C^{ds\_k}_{final}, \tag{8}
$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function and e_{final} represents the output feature of this module. Through this mechanism, the EF module enables the network to more effectively focus on salient features within the target region while suppressing distracting information, thereby enhancing the discriminative capability of the segmentation output.

3.4. Cross-branch semantic supplement module

Owing to the substantial differences in feature processing strategies and representational emphasis across branches, directly fusing features via simple concatenation or element-wise addition often fails to produce truly informative representations for the primary branch and may even interfere with its feature modeling. To address this issue, we propose a CSS module, as illustrated in Fig. 3(e).

First, the global feature C_{gl} aggregated by the GFA module and the shallow or deep feature C_{ds} aggregated by the CFA module are taken as inputs and fed into two parallel feature extraction paths, each serving a distinct role in subsequent processing. Here, C_{ds} is used as a unified notation for the module input, where d and s denote deep and shallow features, respectively. In practice, the module receives either C_d or C_s at a time, for clarity and simplicity, both are collectively denoted as C_{ds}.

The first path uses a Convolution-GELU-Convolution (CGC) structure to generate attention weights, while the second path adopts a Convolution-GELU-Convolution-GELU (CGCG) structure to extract features for subsequent element-wise fusion. Both structures provide nonlinear transformation for local semantic modeling. Unlike CGCG, the CGC branch removes the final GELU because its output is directly fed into a Sigmoid function to produce attention weights. An additional GELU may disturb the feature distribution and impair attention generation. The computation is formulated as follows:

$$
\begin{cases}
C^1_{ds} = \text{CGC}(C_{ds}),\\
C^1_{gl} = \text{CGC}(C_{gl}),\\
C^2_{ds} = \text{CGCG}(C_{ds}),\\
C^2_{gl} = \text{CGCG}(C_{gl}),
\end{cases}
\tag{9}
$$

where CGC(·) denotes the Convolution-GELU-Convolution structure, while CGCG(·) represents the Convolution-GELU-Convolution-GELU structure. C^1_{ds} and C^1_{gl} correspond to the shallow (or deep) feature and the global feature produced by the CGC branch, respectively, whereas C^2_{ds} and C^2_{gl} denote the shallow (or deep) feature and the global feature generated by the CGCG branch. Subsequently, the two features output by the CGC branch are individually passed through a Sigmoid activation function to generate the corresponding attention weights, which can be formulated as follows:

$$
\begin{cases}
C^{att1}_{ds} = \sigma(C^1_{ds}),\\
C^{att1}_{gl} = \sigma(C^1_{gl}),
\end{cases}
\tag{10}
$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, and C^{att1}_{ds} and C^{att1}_{gl} represent the attention weights of the shallow (or deep) feature and the global feature, respectively, generated by the CGC branch followed by Sigmoid activation. Notably, these attention weights are not applied to their own branch, but are used to modulate the input features of the opposite path. Specifically, C^{att1}_{ds} performs element-wise weighting on the initial features of the global branch, while C^{att1}_{gl} modulates the initial features of the shallow (or deep) branch. The attention-modulated features are then element-wise added to the corresponding CGCG outputs from the opposite path, preserving branch-specific information while integrating complementary features. This process enhances multi-level semantic representations and is formulated as follows:

$$
\begin{cases}
C'_{dsgl} = C^{att1}_{ds} \times C_{gl} + C^2_{ds},\\
C'_{glds} = C^{att1}_{gl} \times C_{ds} + C^2_{gl},
\end{cases}
\tag{11}
$$

where C'_{dsgl} and C'_{glds} denote the feature representations obtained after cross-branch attention modulation and subsequent fusion with their respective path features. These two feature streams are then further integrated and refined through additional feature modeling. Specifically, the fused features first undergo a feature extraction operation to consolidate the complementary information introduced during cross-branch modulation. Next, the resulting features are mapped through a Sigmoid function to generate attention weights:

$$
C_{all} = \sigma(\text{CBGC}(C'_{glds} + C'_{dsgl})), \tag{12}
$$

where CBGC(·) denotes the Convolution-BatchNorm-GELU-Convolution structure, in which batch normalization is used to reduce distribution differences between branches and stabilize attention generation. C_{all} represents the attention weights obtained by applying CBGC(·) to the fused features C'_{dsgl} and C'_{glds}.

Based on C_{all}, the network adopts a complementary weighting strategy for the two initial feature paths: one path is element-wise multiplied by C_{all} to emphasize more discriminative features, while the other is multiplied by $1 - C_{all}$ to preserve complementary information. The two weighted features are then fused by element-wise addition to produce an intermediate feature for subsequent processing:

$$
C'_{all} = C'_{glds} \times C_{all} + C'_{dsgl} \times (1 - C_{all}), \tag{13}
$$

where C'_{all} denotes the enhanced feature obtained by summing the two weighted features. To further stabilize its distribution, it is fed into a Convolution-BatchNorm-GELU layer to produce the final output $C^{ds\_k}_{final}$, where $k \in 1,2,3,4$. This feature serves as the result of cross-branch supplementation and is directly applied to the shallow and deep features of the PVT branch, while the global feature of the convolutional branch remains unchanged, thereby completing the cross-branch semantic enhancement.

3.5. Loss function

In this study, for the binary segmentation task, the weighted Intersection over Union (wIoU) loss and the weighted Binary Cross-Entropy (wBCE) loss are adopted as the optimization objectives. Following the deep supervision strategy, this loss is applied to all four segmentation outputs $\{Output_1, Output_2, Output_3, Output_4\}$:

$$
L_{\text{bSeg}} = \sum_{i=1}^{4} \left(\lambda \cdot L^{(i)}_{\text{wIoU}} + \mu \cdot L^{(i)}_{\text{wBCE}} \right), \tag{14}
$$

where λ and μ are weighting coefficients. For the multi-class segmentation task, Dice loss and cross-entropy (CE) loss are jointly adopted as the optimization objectives, and a deep supervision strategy is equally applied to all four segmentation outputs:

$$L_{\text{mSeg}} = \sum_{i=1}^{4} \left(\alpha \cdot L_{\text{Dice}}^{(i)} + \beta \cdot L_{\text{CE}}^{(i)} \right), \tag{15}$$

where α and β are weighting coefficients. In this study, the same coefficient settings as those used in EMCAD (Rahman et al., 2024) are adopted, where $\alpha = 0.7$ and $\beta = 0.3$.

4. Experiments

To comprehensively evaluate the effectiveness and generalization capability of the proposed model, extensive experiments were conducted on six publicly available medical image segmentation datasets.

4.1. Dataset

ISIC 2017 (Codella et al., 2018): This dataset comprises 2000 training images, 150 validation images, and 600 testing images.

ISIC 2018 (Codella et al., 2019): The dataset includes 2594 training images, 100 for validation, and 1000 for testing.

BUSI (Al-Dhabyani et al., 2020): Following the experimental protocol outlined in (Valanarasu & Patel, 2022), a total of 647 images were selected from this dataset, and partitioned into training, validation, and testing sets in an 8:1:1 ratio using the EMCAD (Rahman et al., 2024) methodology.

DSB18 (Caicedo et al., 2019): Containing 670 images in total, this dataset was similarly divided into training, validation, and testing subsets in an 8:1:1 ratio according to the EMCAD (Rahman et al., 2024) strategy.

Synapse: The Synapse[1] multi-organ dataset contains 30 abdominal CT scans, totaling 3779 axial contrast-enhanced slices. Following the approach of EMCAD (Rahman et al., 2024), 18 scans (2212 axial slices) are used for training, and 12 scans are used for testing. The main task is to segment eight abdominal organs: the aorta, gallbladder, left kidney, right kidney, liver, pancreas, spleen, and stomach.

ACDC: The ACDC[2] dataset contains 100 cardiac MRI scans, covering three organs: the right ventricle, myocardium, and left ventricle. We follow the same splitting strategy as EMCAD (Rahman et al., 2024), using 70 scans (1930 axial slices) for training, 10 scans for validation, and 20 scans for testing.

4.2. Evaluation metrics

In this study, we use the Intersection over Union (IoU) and Dice coefficient as the primary performance evaluation metrics, with the corresponding formulas given below.

$$\text{Dice} = \frac{2 \cdot TP}{(TP + FP) + (TN + FN)}, \tag{16}$$

$$\text{IoU} = \frac{TP}{TP + FN + FP}, \tag{17}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. Specifically, TP refers to the number of pixels correctly identified by the model as belonging to the target region, TN represents the number of pixels accurately recognized as background, FP indicates the number of background pixels incorrectly classified as part of the target region, and FN refers to the number of target region pixels mistakenly identified as background.

[1] https://www.synapse.org/Synapse:syn3193805/wiki/217789
[2] https://www.creatis.insa-lyon.fr/Challenge/acdc/

4.3. Experiment details

All experiments were conducted on a server equipped with an NVIDIA RTX A800 GPU. For binary segmentation tasks, all input images were first uniformly resized to 256×256 pixels, and the same data augmentation strategy as that used in TransFuse (Zhang et al., 2021) was adopted during training. More specifically, the model was trained for 30 epochs on both the ISIC 2017 and ISIC 2018 datasets with an initial learning rate of 1e-4; on the BUSI dataset, training was performed for 50 epochs with an initial learning rate of 5e-5; for the DSB18 dataset, the training duration was extended to 200 epochs, with the learning rate likewise set to 5e-5. For multi-class segmentation tasks, a different experimental configuration was employed. The Synapse dataset was trained for 500 epochs, whereas the ACDC dataset was trained for 100 epochs, with images from both datasets resized to 224×224 pixels. Under this setting, the batch size and learning rate were fixed at 6 and 1e-4, respectively. It should also be noted that AdamW was consistently used as the optimizer in all experiments.

4.4. Ablation study

To assess the contribution of each module to the model's performance, we conducted systematic ablation experiments across multiple datasets. The Table 1 presents segmentation results for various combinations of the CFA, CSS, GFA, and EF modules, including IoU and Dice metrics. The results indicate that employing the CFA module alone already achieves strong performance across datasets-for instance, the Dice score on ISIC 2017 reaches 85.81%. Adding the EF module further improves performance, demonstrating its ability to enhance the discriminative power of feature representations.

Incorporating the CSS and GFA modules leads to additional gains; for example, on the BUSI dataset, the IoU improves from 72.07% to 74.45%, highlighting the effectiveness of cross-branch semantic supplementation and global feature enhancement in capturing critical information. When all four modules are applied simultaneously, the model attains optimal performance across all datasets-for instance, Dice scores rise to 90.63% on ISIC 2018, 91.94% on DSB18, 84.32% on Synapse, and 92.40% on ACDC. These results indicate that the modules work synergistically to reinforce cross-feature semantic consistency, supplement complementary information, and strengthen global feature responses, thereby significantly improving semantic representation and segmentation accuracy. Overall, the performance consistently surpasses that of partial module combinations, validating the effectiveness and robustness of the proposed module design.

4.5. Comparative experiments of CFA and CSS with traditional feature fusion methods

We conducted independent comparative experiments on the CFA and CSS modules using the ISIC 2017 dataset. Specifically, two traditional feature fusion strategies-element-wise addition (Addition) and channel-wise concatenation (Concatenation)-were employed to replace each module individually. Table 2 summarizes the performance results obtained when either module was replaced separately as well as when both modules were replaced simultaneously. The experimental results show that, under the single-module replacement setting, "Concatenation (replacing CFA)" achieves the strongest performance, with IoU and Dice scores of 79.23% and 87.43%, respectively, ranking just below the complete model. By contrast, when both modules are replaced at the same time, the model performance declines noticeably, with "Concatenation (replacing CFA) + Addition (replacing CSS)" yielding the worst results. For comparison, the full CC-Net attains the best performance across all metrics, achieving an IoU of 81.04% and a Dice score of 88.47%. These findings demonstrate the effectiveness of the proposed CFA and CSS modules in improving segmentation performance.

Table 1
Ablation study results on different datasets.

CFA	CSS	GFA	EF	ISIC 2017		ISIC 2018		BUSI		DSB18		Synapse	ACDC
				IoU(%)↑	Dice(%)↑	IoU(%)↑	Dice(%)↑	IoU(%)↑	Dice(%)↑	IoU(%)↑	Dice(%)↑	Dice(%)↑	Dice(%)↑
✓				77.93	85.81	80.92	87.90	72.07	80.15	82.32	89.17	81.78	89.62
✓			✓	78.68	86.77	81.74	88.82	72.71	80.99	83.12	90.11	82.64	90.56
✓	✓			78.33	86.32	81.36	88.46	72.42	80.67	82.77	89.77	82.59	90.37
	✓		✓	78.13	85.97	81.14	88.27	72.26	80.44	82.53	89.39	82.33	90.33
	✓	✓		78.98	86.95	82.02	89.07	72.99	81.22	83.44	90.36	82.87	90.81
	✓	✓	✓	79.34	87.13	82.86	89.21	72.63	81.07	84.57	91.22	82.76	91.27
✓	✓	✓		80.57	87.86	83.67	90.12	74.45	82.23	85.11	91.31	83.74	91.88
✓	✓		✓	79.92	87.36	83.31	89.62	73.33	81.56	84.83	91.27	83.12	91.73
✓	✓	✓	✓	81.04	88.47	84.16	90.63	74.89	82.64	85.61	91.94	84.32	92.40
Mean				79.21	87.18	82.24	89.23	73.08	81.22	83.92	90.51	82.91	91.00
Standard Deviation				1.15	0.84	1.29	0.91	1.08	0.78	1.17	0.98	0.75	0.84

Table 2
Comparative experimental results of cfa and css with traditional feature fusion methods on the isic 2017 dataset. the best-performing model is highlighted in bold, and the second-best is underlined.

Different Method Combinations	IoU(%)↑	Dice(%)↑
Addition (Replacing CFA)	78.93	<u>87.53</u>
Concatenation (Replacing CFA)	79.23	87.43
Addition (Replacing CSS)	79.14	87.31
Concatenation (Replacing CSS)	<u>79.32</u>	87.32
Addition (Replacing CFA) + Addition (Replacing CSS)	78.41	85.88
Addition (Replacing CFA) + Concatenation (Replacing CSS)	76.29	85.53
Concatenation (Replacing CFA) + Addition (Replacing CSS)	76.16	85.46
Concatenation (Replacing CFA) + Concatenation (Replacing CSS)	77.05	86.21
CFA + CSS (Ours)	**81.04**	**88.47**

Table 3
Comparative experimental results for binary segmentation tasks across the ISIC 2017, ISIC 2018, BUSI, and DSB18 datasets. In each column, the best-performing results are highlighted in bold, while the second-best results are indicated with underlining.

Method	ISIC 2017		ISIC 2018		BUSI		DSB18	
	IoU(%)↑	Dice(%)↑	IoU(%)↑	Dice(%)↑	IoU(%)↑	Dice(%)↑	IoU(%)↑	Dice(%)↑
U-Net (Ronneberger et al., 2015)	72.34	81.59	77.33	85.45	50.70	67.29	81.42	88.16
PraNet (Fan et al., 2020)	76.96	85.28	82.12	88.93	71.18	79.61	84.75	91.40
TransFuse (Zhang et al., 2021)	79.50	87.20	82.56	89.33	73.46	81.45	83.99	90.74
UNeXt (Valanarasu & Patel, 2022)	74.06	82.51	78.42	87.36	59.19	74.28	83.67	90.89
CASCADE (Rahman & Marculescu, 2023)	77.80	85.89	81.06	88.46	72.65	80.72	84.82	91.54
EMCAD (Rahman et al., 2024)	78.66	86.58	81.93	89.20	72.67	81.41	<u>85.11</u>	<u>91.58</u>
CPSR (Yin et al., 2024b)	73.76	83.12	79.53	87.28	59.45	70.95	80.67	88.78
MDRL (Yin et al., 2024a)	75.20	84.10	79.04	87.34	62.24	72.72	81.15	88.84
ProtoSeg (Zhou & Wang, 2024)	76.97	85.41	81.85	88.67	63.90	73.92	82.14	90.23
MCCL (Yin et al., 2025)	74.82	83.45	79.56	87.50	61.50	71.84	80.70	89.01
VM-Unet (Ruan et al., 2025)	78.92	86.53	82.21	89.86	72.23	79.95	83.56	90.97
CC-Net-b0(Ours)	<u>80.18</u>	<u>87.78</u>	<u>82.84</u>	<u>90.16</u>	<u>73.57</u>	<u>81.88</u>	84.85	91.42
CC-Net-b2(Ours)	**81.04**	**88.47**	**84.16**	**90.63**	**74.89**	**82.64**	**85.61**	**91.94**

4.6. Results of binary medical image segmentation

Table 3 presents the comparative experimental results on four binary medical image segmentation datasets (ISIC 2017, ISIC 2018, BUSI, and DSB18). The proposed CC-Net-b2 achieves the best performance across all datasets. Fig. 4 illustrates the corresponding qualitative results, where the first column shows the original images, the second column presents the ground truth annotations, columns three to six provide comparisons with several state-of-the-art methods, and the final column displays the predictions of the proposed model. In the visualization, red contours denote the ground truth boundaries, while green contours indicate the predicted segmentation boundaries.

ISIC 2017 and ISIC 2018 Datasets: On the ISIC 2017 dataset, our CC-Net-b2 achieves an IoU of 81.04% and a Dice score of 88.47%, while on ISIC 2018, it attains an IoU of 84.16% and a Dice score of 90.63%. Compared with benchmark models, CC-Net demonstrates notable improvements: Dice increases by +1.89% and +1.43% over EMCAD, +3.24% and +2.17% over CASCADE, and +5.96% and +3.27% over UNeXt, respectively. These results indicate that CC-Net more accurately captures lesion boundaries and overall structure in skin lesion segmentation tasks.

BUSI Datasets: On the BUSI dataset, CC-Net-b2 achieves an IoU of 74.89% and a Dice score of 82.64%. Compared with benchmark models, Dice and IoU improve by +1.19% and +1.43% over TransFuse, +11.69% and +14.44% over CPSR, and +9.92% and +12.65% over MDRL, respectively. These results indicate that CC-Net more accurately delineates lesion regions in breast ultrasound images while effectively reducing background false positives.

DSB18 Datasets: On the DSB18 dataset, CC-Net-b2 achieves an IoU of 85.61% and a Dice score of 91.94%. Compared with benchmark models, it improves Dice and IoU by +0.36% and +0.50% over EMCAD, +1.71% and +3.47% over ProtoSeg, and +2.93% and +4.91% over

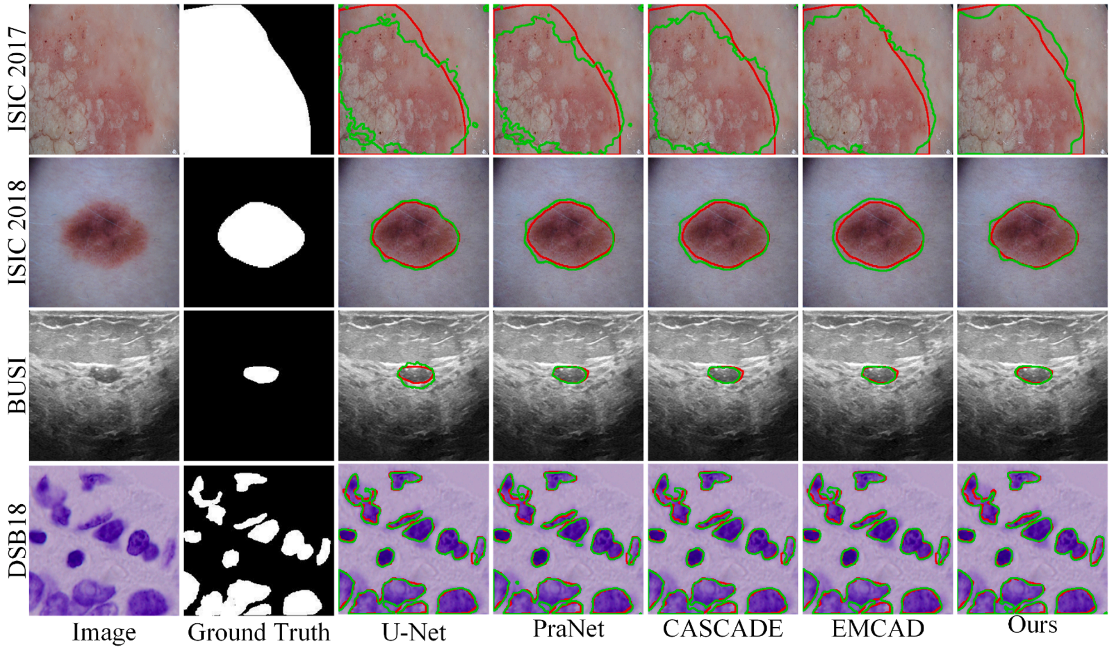

Fig. 4. The visualization results of binary medical image segmentation show red contours representing the annotated ground truth boundaries and green contours representing the predicted segmentation boundaries.

Table 4
Comparative experimental results for multi-class segmentation tasks on the Synapse and ACDC datasets. In each column, the best-performing results are highlighted in bold, while the second-best results are indicated with underlining.

Method	Synapse	ACDC
	Dice(%)↑	Dice(%)↑
U-Net (Ronneberger et al., 2015)	76.85	90.34
MISSFormer (Huang et al., 2022)	81.96	87.73
CASCADE (Rahman & Marculescu, 2023)	82.68	91.46
EMCAD (Rahman et al., 2024)	<u>83.63</u>	92.12
Swin-UMamba (Liu et al., 2024b)	77.42	<u>92.24</u>
CPSR (Yin et al., 2024b)	75.62	90.48
MDRL (Yin et al., 2024a)	75.71	90.45
ProtoSeg (Zhou & Wang, 2024)	78.72	90.77
MCCL (Yin et al., 2025)	76.67	90.07
VM-Unet (Ruan et al., 2025)	81.08	89.58
Zig-RiR (Chen et al., 2025)	74.61	87.42
CC-Net-b0(Ours)	83.51	91.87
CC-Net-b2(Ours)	**84.32**	**92.40**

Table 5
Ablation study results of different component combinations of the ef module on the isic 2017 dataset. the best-performing model is highlighted in bold, and the second-best is underlined.

GAP	GMP	CBG	IoU (%)↑	Dice(%)↑
✓			79.78	88.09
	✓		79.82	88.05
		✓	80.26	88.08
✓	✓		80.74	88.24
✓		✓	<u>80.86</u>	88.27
	✓	✓	80.58	<u>88.35</u>
✓	✓	✓	**81.04**	**88.47**

MCCL, respectively. These results indicate that CC-Net can consistently and accurately segment nuclei of varying shapes and sizes, while enhancing feature discriminability and overall robustness.

4.7. Results of multi-class medical image segmentation

Table 4 presents the comparative results on two multi-class medical image segmentation datasets (Synapse and ACDC). The proposed CC-Net-b2 achieves the highest performance across all datasets. Fig. 5 provides the corresponding qualitative visualizations. The first column shows the ground truth annotations, columns two to five display comparisons with several state-of-the-art methods, and the final column presents the predictions of the proposed model. Specifically, the first row depicts 2D visualizations for the Synapse dataset, the second row shows 3D visualizations for Synapse, the third row illustrates 2D visualizations for the ACDC dataset, and the fourth row presents 3D visualizations for ACDC.

Synapse Datasets: On the Synapse dataset, our CC-Net-b2 achieves a Dice score of 84.32%. Compared with the benchmark models, it demonstrates clear improvements: Dice increases by 0.69% over EMCAD, by 6.90% over Swin-UMamba, by 3.24% over VM-Unet, and by 1.64% over CASCADE. These results indicate that CC-Net more accurately segments abdominal organ regions, thereby enhancing both segmentation precision and robustness.

ACDC Datasets: On the ACDC dataset, CC-Net-b2 achieves a Dice score of 92.40%. Compared with benchmark models, it demonstrates improvements of +0.28% over EMCAD, +0.16% over Swin-UMamba, +0.94% over CASCADE, and +0.82% over VM-Unet. These results indicate that CC-Net more precisely captures cardiac structure boundaries while maintaining high overall consistency and discriminative capability.

4.8. Ablation study on different component combinations of the EF module

Table 5 reports the comparative experimental results of the EF module and its internal components, namely GAP, GMP, and CBG, on the ISIC 2017 dataset. From the overall trend, the inclusion of each individual component leads to a certain degree of performance improvement, among which CBG performs noticeably better than both GAP and GMP. A closer examination of different component combinations further

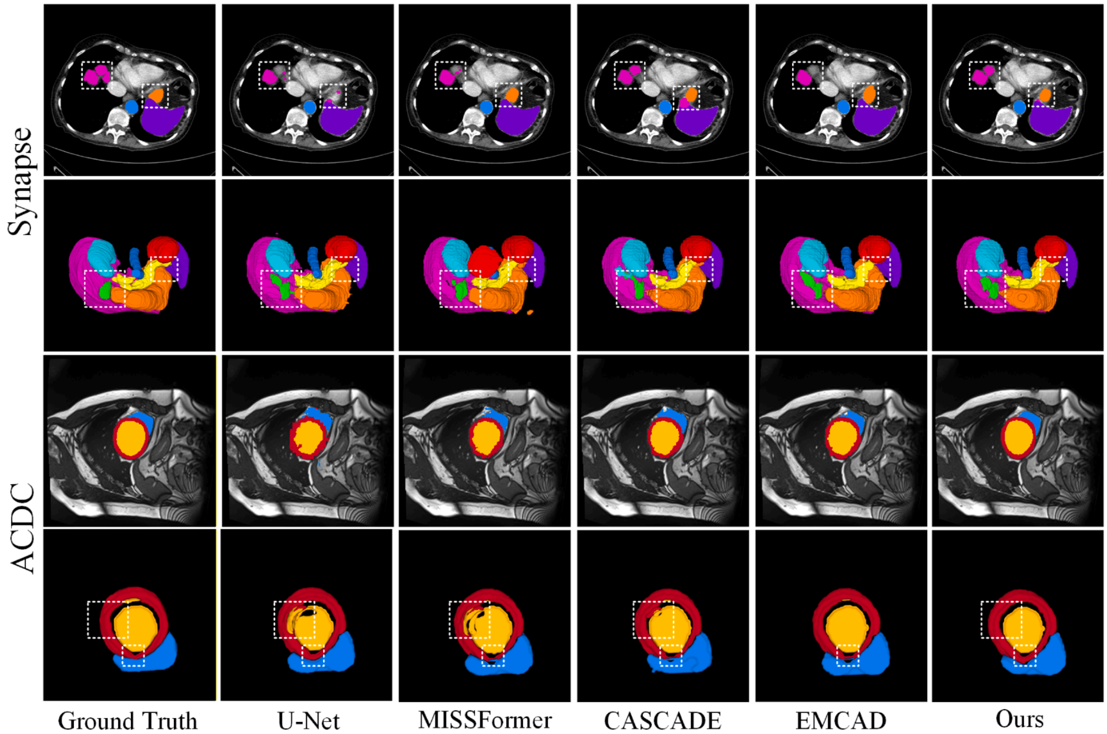

Fig. 5. Visualization results of multi-class medical image segmentation. First row: 2D visualizations of the Synapse dataset; second row: 3D visualizations of the Synapse dataset; third row: 2D visualizations of the ACDC dataset; fourth row: 3D visualizations of the ACDC dataset.

Table 6
Sensitivity analysis of loss function weights on the ISIC 2017 Dataset.

λ	μ	IoU(%)↑	Dice(%)↑
0.2	0.8	78.27	85.37
0.3	0.7	78.61	85.82
0.4	0.6	79.01	86.26
0.6	0.4	79.82	87.14
0.7	0.3	79.42	86.70
0.8	0.2	79.66	86.97
1.0	**1.0**	**81.04**	**88.47**

Table 7
Quantitative comparison of model parameters and flops across different models.

Model	Params(M)	FLOPs(G)
U-Net (Ronneberger et al., 2015)	13.40	58.83
PraNet (Fan et al., 2020)	30.50	13.15
TransFuse (Zhang et al., 2021)	143.74	82.71
UNeXt (Valanarasu & Patel, 2022)	1.47	9.17
MISSFormer (Huang et al., 2022)	35.45	7.28
CASCADE (Rahman & Marculescu, 2023)	34.12	7.62
EMCAD (Rahman et al., 2024)	26.77	5.60
Swin-UMamba (Liu et al., 2024b)	59.89	43.94
VM-Unet (Ruan et al., 2025)	22.05	4.12
Zig-RiR (Chen et al., 2025)	24.58	6.23
CC-Net-b0(Ours)	8.63	69.45
CC-Net-b2(Ours)	30.10	73.85

reveals that combining GAP with GMP improves model performance relative to using either component alone, whereas pairing CBG with either GAP or GMP yields even more pronounced gains. Among these settings, the "GMP + CBG" configuration achieves the second-highest Dice score. Ultimately, when all three components are incorporated simultaneously, that is, under the "GAP + GMP + CBG" configuration, the model attains the best overall performance, with IoU and Dice reaching 81.04% and 88.47%, respectively. These results demonstrate that the individual components of the EF module are not only effective in their own right, but also exhibit strong complementarity when combined.

4.9. Sensitivity analysis of loss function weights

To evaluate the impact of the loss function hyperparameters λ and μ on the performance of binary medical image segmentation, we conducted a sensitivity analysis on the validation set of the ISIC 2017 dataset and reported the corresponding performance on the test set under different parameter combinations (Table 6). The results indicate that the model achieves optimal IoU and Dice scores when λ and μ are approximately balanced, whereas deviations from this balance lead to performance degradation. This demonstrates that the weighted IoU loss and weighted binary cross-entropy loss contribute equally to the training process.

4.10. Comparison of model parameters and FLOPs

We conducted a complexity comparison among different U-shaped architectures, with results summarized in Table 7, using model parameters and floating-point operations as evaluation metrics. CC-Net-b0 has only 8.63M parameters, while CC-Net-b2 has 30.10M, both lower than several mainstream models such as TransFuse (143.74M), MISSFormer (35.45M), and CASCADE (34.12M), indicating that the multi-module design does not substantially inflate model size. The FLOPs are 69.45G and 73.85G, respectively, which are reasonable compared with U-Net (58.83G) and TransFuse (82.71G), yet higher than lightweight models such as UNeXt, EMCAD, and VM-Unet, primarily due to the additional computations required for cross-hierarchical feature modeling. Overall, CC-Net achieves a balanced trade-off between performance enhancement and computational complexity, effectively strengthening the representation of complex structures and fine-grained regions within a controlled parameter budget.

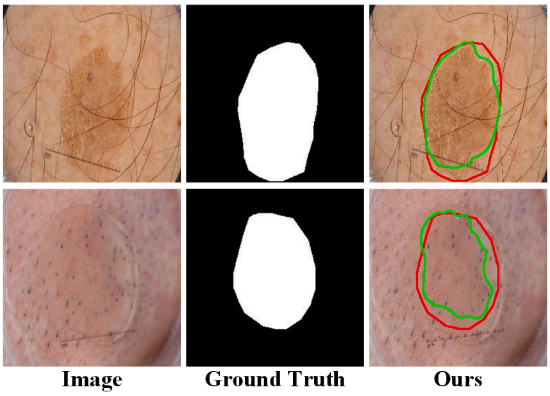

Fig. 6. Examples of incomplete or inaccurate segmentation caused by low-level visual feature similarity.

5. Limitations and future work

Although CC-Net achieves significant improvements in segmentation performance, it still has certain limitations. First, when the target regions and the background exhibit high similarity in low-level visual features such as color and texture, the model may struggle to effectively distinguish the target areas. As shown in Fig. 6, in some cases, due to the lack of apparent differences in appearance between the target and background, the model's discriminative capability is limited, leading to incomplete or inaccurate identification of the target regions. Second, CC-Net does not fully address interpolation issues caused by resolution discrepancies during cross-layer feature fusion. Since features from different layers vary considerably in spatial resolution and semantic hierarchy, cross-layer fusion typically relies on upsampling or downsampling operations, which may introduce unnecessary smoothing or structural distortions, resulting in artifacts-particularly along fine-grained boundaries. Additionally, the multi-module design of CC-Net, while enhancing segmentation accuracy, incurs increased computational overhead, which may limit its applicability in real-time scenarios or resource-constrained environments.

To address these limitations, future work will focus on improving both model performance and practical usability. On one hand, more precise cross-scale feature alignment and fusion strategies will be explored to bridge the semantic gap between features of different resolutions, reduce artifacts, and enhance boundary modeling. On the other hand, more efficient architectural designs or lightweight strategies will be introduced to maintain segmentation accuracy while reducing computational complexity, thereby improving the model's potential for clinical and engineering applications.

6. Conclusions

This study proposes the CC-Net, designed to overcome the limitations of conventional U-shaped architectures in cross-level feature interaction. By explicitly modeling semantic relationships between shallow and deep features through the CFA module, constructing a unified global semantic representation via the GFA module, supplementing layer-specific features with global context through the CSS module, and enhancing target region discrimination with the EF module, CC-Net demonstrates superior capability in delineating complex boundaries and fine-grained structures. Extensive experiments across six publicly available medical imaging datasets show that CC-Net consistently achieves outstanding segmentation performance, outperforming existing state-of-the-art methods across multiple evaluation metrics. In summary, CC-Net not only advances cross-level feature fusion and semantic awareness but also provides an effective solution for precise lesion segmentation in medical images, laying a solid foundation for subsequent clinical applications.

CRediT authorship contribution statement

Xiaoyan Zhang: Writing – review & editing, Supervision; **Zheng Zhao:** Writing – review & editing, Writing – original draft, Methodology; **Weiqiang Sun:** Writing – review & editing; **Yongqin Zhang:** Writing – review & editing, Methodology; **Chunlin Yu:** Writing – review & editing, Methodology; **Xiangfu Meng:** Writing – review & editing, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Liaoning Provincial Natural Science Foundation Joint Program Project (20240311).

References

Al-Dhabyani, W., Gomaa, M., Khaled, H., & Fahmy, A. (2020). Dataset of breast ultrasound images. *Data in brief*, *28*, 104863.

Azad, R., Aghdam, E. K., Rauland, A., Jia, Y., Avval, A. H., Bozorgpour, A., Karimijafarbigloo, S., Cohen, J. P., Adeli, E., & Merhof, D. (2024). Medical image segmentation review: The success of u-net. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *46*, 10076–10095.

Caicedo, J. C., Goodman, A., Karhohs, K. W., Cimini, B. A., Ackerman, J., Haghighi, M., Heng, C., Becker, T., Doan, M., McQuin, C. et al. (2019). Nucleus segmentation across imaging experiments: The 2018 data science bowl. *Nature methods*, *16*(12), 1247–1253.

Cao, H., Wang, Y., Chen, J., Jiang, D., Zhang, X., Tian, Q., & Wang, M. (2022). Swin-unet: Unet-like pure transformer for medical image segmentation. In *European conference on computer vision*, pages 205–218. Springer.

Chen, G., Li, L., Dai, Y., Zhang, J., & Yap, M. H. (2022). Aau-net: An adaptive attention u-net for breast lesions segmentation in ultrasound images. *IEEE Transactions on Medical Imaging*, *42*(5), 1289–1300.

Chen, T., Zhou, X., Tan, Z., Wu, Y., Wang, Z., Ye, Z., Gong, T., Chu, Q., Yu, N., & Lu, L. (2025). Zig-rir: Zigzag rwkv-in-rwkv for efficient medical image segmentation. *IEEE Transactions on Medical Imaging*, *44*, 3245–3257.

Codella, N., Rotemberg, V., Tschandl, P., Celebi, M. E., Dusza, S., Gutman, D., Helba, B., Kalloo, A., Liopyris, K., Marchetti, M. et al. (2019). Skin lesion analysis toward melanoma detection 2018: a challenge hosted by the international skin imaging collaboration (isic). arXiv preprint arXiv:1902.03368.

Codella, N. C. F., Gutman, D., Celebi, M. E., Helba, B., Marchetti, M. A., Dusza, S. W., Kalloo, A., Liopyris, K., Mishra, N., Kittler, H. et al. (2018). Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic). In *2018 IEEE 15Th international symposium on biomedical imaging (ISBI 2018)*, pages 168–172. IEEE.

Dai, D., Dong, C., Yan, Q., Sun, Y., Zhang, C., Li, Z., & Xu, S. (2024). I2u-Net: A dual-path u-net with rich information interaction for medical image segmentation. *Medical Image Analysis*, *97*, 103241.

Ding, Y., Li, L., Wang, W., & Yang, Y. (2024). Clustering propagation for universal medical image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3357–3369.

Fan, D.-P., Ji, G.-P., Zhou, T., Chen, G., Fu, H., Shen, J., & Shao, L. (2020). Pranet: Parallel reverse attention network for polyp segmentation. In *International conference on medical image computing and computer-assisted intervention*, pages 263–273. Springer.

Huang, H., Lin, L., Tong, R., Hu, H., Zhang, Q., Iwamoto, Y., Han, X., Chen, Y.-W., & Wu, J. (2020). Unet 3+: A full-scale connected unet for medical image segmentation. In *ICASSP 2020-2020 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 1055–1059. IEEE.

Huang, S., Ge, Y., Liu, D., Hong, M., Zhao, J., & Loui, A. C. (2025). Rethinking copy-paste for consistency learning in medical image segmentation. *IEEE Transactions on Image Processing*, *34*, 1060–1074.

Huang, X., Deng, Z., Li, D., Yuan, X., & Fu, Y. (2022). Missformer: An effective transformer for 2d medical image segmentation. *IEEE Transactions On Medical Imaging*, *42*(5), 1484–1494.

Ibtehaz, N., & Kihara, D. (2023). Acc-unet: A completely convolutional unet model for the 2020s. In *International conference on medical image computing and computer-assisted intervention*, pages 692–702. Springer.

Isensee, F., Jaeger, P. F., Kohl, S. A. A., Petersen, J., & Maier-Hein, K. H. (2021). Nnu-net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature methods*, *18*(2), 203–211.

Jha, D., Riegler, M. A., Johansen, D., Halvorsen, P., & Johansen, H. D. (2020). Doubleu-net: A deep convolutional neural network for medical image segmentation. In *2020 IEEE 33Rd international symposium on computer-based medical systems (CBMS)*, pages 558–564. IEEE.

Jiang, J., Zhang, J., Liu, W., Gao, M., Hu, X., Yan, X., Huang, F., & Liu, Y. (2025). Rwkv-unet: Improving unet with long-range cooperation for effective medical image segmentation. arXiv preprint arXiv:2501.08458.

Kuang, H., Wang, Y., Tan, X., Yang, J., Sun, J., Liu, J., Qiu, W., Zhang, J., Zhang, J., Yang, C. et al. (2025). Lw-ctrans: A lightweight hybrid network of cnn and transformer for 3d medical image segmentation. *Medical Image Analysis*, *102*, 103545.

Liu, J., Li, K., Huang, C., Dong, H., Song, Y., & Li, R. (2024a). Mixformer: A mixed CNN-transformer backbone for medical image segmentation. *IEEE Transactions on Instrumentation and Measurement*, *74*, 5001220.

Liu, J., Yang, H., Zhou, H.-Y., Yu, L., Liang, Y., Yu, Y., Zhang, S., Zheng, H., & Wang, S. (2024b). Swin-UMamba†: Adapting mamba-based vision foundation models for medical image segmentation. *IEEE Transactions on Medical Imaging*, *44*, 3898–3908.

Liu, Y., Feng, Y., Cheng, J., Zhan, H., & Zhu, Z. (2025). Mambadiff: Mamba-enhanced diffusion model for 3d medical image segmentation. *IEEE Transactions on Image Processing*, *34*, 5761–5775.

Long, J., Shelhamer, E., & Darrell, T. (2015). Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3431–3440.

Lu, Y., Yang, Y., Xing, Z., Wang, Q., & Zhu, L. (2024). Diff-vps: Video polyp segmentation via a multi-task diffusion network with adversarial temporal reasoning. In *International conference on medical image computing and computer-assisted intervention*, pages 165–175. Springer.

Lumetti, L., Pipoli, V., Marchesini, K., Ficarra, E., Grana, C., & Bolelli, F. (2025). Taming mambas for 3d medical image segmentation. *IEEE Access*, *13*, 89748–89759.

Rahman, M. M., & Marculescu, R. (2023). Medical image segmentation via cascaded attention decoding. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 6222–6231.

Rahman, M. M., Munir, M., & Marculescu, R. (2024). Emcad: Efficient multi-scale convolutional attention decoding for medical image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11769–11779.

Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, munich, germany, october 5–9, 2015, proceedings, part III 18*, pages 234–241. Springer.

Ruan, J., Li, J., & Xiang, S. (2025). Vm-unet: Vision mamba unet for medical image segmentation. *ACM Transactions on Multimedia Computing, Communications and Applications*. https://doi.org/10.1145/3767748.

Shao, H., Zeng, Q., Hou, Q., & Yang, J. (2025). Mcanet: Medical image segmentation with multi-scale cross-axis attention. *Machine Intelligence Research*, *22*(3), 437–451.

Valanarasu, J. M. J., Oza, P., Hacihaliloglu, I., & Patel, V. M. (2021). Medical transformer: Gated axial-attention for medical image segmentation. In *Medical image computing and computer assisted intervention–MICCAI 2021: 24th international conference, strasbourg, france, september 27–october 1, 2021, proceedings, part i 24*, pages 36–46. Springer.

Valanarasu, J. M. J., & Patel, V. M. (2022). Unext: Mlp-based rapid medical image segmentation network. In *International conference on medical image computing and computer-assisted intervention*, pages 23–33. Springer.

Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., & Shao, L. (2021). Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 568–578.

Wu, T., Li, B., Luo, Y., Wang, Y., Xiao, C., Liu, T., Yang, J., An, W., & Guo, Y. (2023). Mtu-net: Multilevel transunet for space-based infrared tiny ship detection. *IEEE Transactions on Geoscience and Remote Sensing*, *61*, 1–15.

Yin, J., Chen, T., Pei, G., Liu, H., Yao, Y., Nie, L., & Hua, X. (2025). Semi-supervised semantic segmentation with multi-constraint consistency learning. *IEEE Transactions on Multimedia*, *27*, 6449–6461.

Yin, J., Peng, N., Chen, Y., Zheng, Z., Gu, Y., & Zhou, J. (2024a). Class-level multiple distributions representation are necessary for semantic segmentation. In *International conference on database systems for advanced applications*, pages 340–351. Springer.

Yin, J., Yan, S., Chen, T., Chen, Y., & Yao, Y. (2024b). Class probability space regularization for semi-supervised semantic segmentation. *Computer Vision and Image Understanding*, *249*, 104146.

Zeiler, M. D., & Fergus, R. (2014). Visualizing and understanding convolutional networks. In *European conference on computer vision*, pages 818–833. Springer.

Zhang, X., Zhao, Z., Zhang, Y., Yu, C., Meng, X., & Li, S. (2026). Effm: Rethinking feature fusion mechanisms for medical image segmentation. *Biomedical Signal Processing and Control*, *115*, 109334.

Zhang, Y., Liu, H., & Hu, Q. (2021). Transfuse: Fusing transformers and cnns for medical image segmentation. In *Medical image computing and computer assisted intervention–MICCAI 2021: 24th international conference, strasbourg, france, september 27–october 1, 2021, proceedings, part i 24*, pages 14–24. Springer.

Zhang, Z., Liu, Q., & Wang, Y. (2018). Road extraction by deep residual u-net. *IEEE Geoscience and Remote Sensing Letters*, *15*(5), 749–753.

Zhou, T., & Wang, W. (2024). Prototype-based semantic segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *46*(10), 6858–6872.

Zhou, Z., Rahman Siddiquee, M. M., Tajbakhsh, N., & Liang, J. (2018). Unet++: A nested u-net architecture for medical image segmentation. In *Deep learning in medical image analysis and multimodal learning for clinical decision support: 4th international workshop, DLMIA 2018, and 8th international workshop, ML-CDS 2018, held in conjunction with MICCAI 2018, granada, spain, september 20, 2018, proceedings 4*, pages 3–11. Springer.

Zhu, Z., He, X., Qi, G., Li, Y., Cong, B., & Liu, Y. (2023). Brain tumor segmentation based on the fusion of deep semantics and edge information in multimodal MRI. *Information Fusion*, *91*, 376–387.

Zhu, Z., Zhang, Z., Qi, G., Li, Y., Yang, P., & Liu, Y. (2025). Probability map-guided network for 3d volumetric medical image segmentation. *IEEE Transactions on Image Processing*, *34*, 7222–7234.